Safe administration of influenza vaccine to patients with egg allergy

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Objectives: Specific recommendations for administering the influenza vaccine to patients with egg allergy are based on limited scientific data. The objectives of this investigation were to determine the safety of a 2-dose administration of an influenza vaccine to patients with egg allergy and to evaluate the usefulness of skin testing with the influenza vaccine before administration.

Study design: In this multicenter clinical trial, clinical histories of egg allergy were confirmed by skin testing with egg and, if possible, by oral challenges with egg. Subjects with egg allergy received the vaccine in 2 doses, 30 minutes apart; the first dose was one tenth and the second dose nine tenths of the recommended dose as determined by age. Subjects without egg allergy were recruited as control subjects and received 1 age-determined dose of the vaccine. Skin prick tests with the influenza vaccine were performed on all subjects.

Results: From 1994 to 1997, 83 subjects with egg allergy and 124 control subjects were evaluated. The content of ovalbumin/ovomucoid was 0.1, 1.2, and 0.02 μg/mL, respectively in the 1994-95, 1995-96, and 1996-97 influenza vaccines. Results of vaccine skin prick tests were positive in 4 subjects with egg allergy and in 1 control subject. All patients with egg allergy tolerated the vaccination protocol without any significant allergic reactions.

Conclusions: These results demonstrate that patients with egg allergy, even those with significant allergic reactions after egg ingestion, can safely receive an influenza vaccine in a 2-dose protocol when the vaccine preparation contains no more than 1.2 μg/mL egg protein. (J Pediatr 1998;133:624-8)

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Influenza vaccines are produced in embryonated eggs, and a previous investigation reported that after commercial processing, these vaccines contain a small amount (1 to 7 μg/mL) of egg protein.[1] The American Academy of Pediatrics[2] currently recommends that patients with severe, anaphylactic reactions after egg ingestion should not receive an influenza vaccine until they have been appropriately skin tested with a diluted preparation of the vaccine. If there is a positive skin test response to the vaccine preparation, either by the epicutaneous or intradermal skin testing method, the influenza vaccine should not be administered in the usual fashion.[2] In clinical situations in which immunization is imperative, multiple graded doses of the influenza vaccine may be carefully administered by trained personnel experienced in the management of anaphylaxis. These specific recommendations, based on limited scientific evidence, have unfortunately promoted confusion in the lay public and medical community.[3-8] As a result, a group of patients may be denied protection provided by this vaccine because of inadequate confirmation of clinical histories of egg allergy and/or nonspecific irritant reactions to vaccine skin tests.

ELISA	Enzyme-linked immunosorbent assay
MMR	Measles-mumps-rubella

Commercially available measles and mumps vaccines are produced by chick embryo fibroblast tissue cultures and do not contain significant amounts of egg cross-reacting proteins.[9] An accumulation of scientific evidence supports the safe, routine administration of measles vaccines in patients allergic to eggs.[9-11] Children with egg allergy, even those with severe hypersensitivity, are at low risk for anaphylactic reactions to these vaccines, singly or in combination (ie, measles-mumps-rubella). Furthermore, skin testing with dilute preparations of the vaccine is not predictive of an allergic reaction to vaccination. Another vaccine component, gelatin, has recently been implicated in immediate hypersensitivity reactions after administration of the MMR vaccine.[12] The current edition of the American Academy of Pediatrics Red Book supports the routine, 1-dose administration of the MMR, measles, or mumps vaccine to children with egg allergy without prior skin testing with the vaccine.[2]

The specific aims of this multicenter investigation were to investigate the safety of a 2-dose administration of this vaccine to a well-defined group of patients with egg allergy, as compared with its routine administration in a group of healthy subjects without egg allergy, and to evaluate the clinical usefulness of puncture skin testing with the influenza vaccine before administration.

METHODS

Study Design

This investigation was a multicenter (n = 10) clinical trial implemented over a 3-year period covering consecutive influenza seasons (1994 through 1997). Each site had one designated investigator and/or coordinator who was responsible for recruiting patients, performing the research protocol, and recording the research data on standardized clinical worksheets. Informed consent from the subject or a parent or guardian was obtained for all participants in the investigation. Approval from the appropriate institutional review board for research in human subjects was obtained from the individual participating investigators.

Patient Recruitment and Management

Patients, aged 6 months and older, who were clinical candidates for receiving the influenza vaccine were screened for egg hypersensitivity with clinical histories and an epicutaneous or puncture skin test with a commercial egg extract. The egg allergy study group included: (1) subjects (n = 27) with a convincing history of anaphylaxis including generalized urticaria, wheezing, laryngeal edema, and/or hypotension after the ingestion of egg or egg products (these patients were not subjected to an oral challenge with egg because of the increased risk of precipitating an anaphylactic reaction during the challenge), (2) patients (n = 31) with a positive clinical history of egg allergy and a positive skin test response to egg, and (3) patients (n = 25) who had a positive response on blinded, oral challenge with egg. If results of the skin test with egg were positive but the clinical history was equivocal for an allergic reaction after egg ingestion, the patient was asked to participate in a blinded oral challenge with egg to confirm a positive clinical history of egg allergy. Patients with positive results on blinded challenge with egg (n = 25) were entered in the study group, and those with negative challenge results were included as control subjects. Control subjects consisted of 3 different groups including those with negative histories of egg allergy and negative skin test responses to egg (n = 109), positive histories of egg allergy and negative skin test responses to egg (n = 13), and equivocal histories of egg allergy and positive skin test response to egg but negative results on blinded, oral challenge with egg (n = 2).

To help determine the ability of a positive skin test response to the influenza vaccine to predict a subsequent adverse reaction after its administration, all enrollees underwent skin prick testing by the epicutaneous method with the full-strength influenza vaccine before immunization. Subjects were excluded from receiving the influenza vaccine if they had: (1) asthma under poor control (forced expiratory volume in 1 second < 70% of usual best effort); (2) acute febrile illnesses (T >101°F); and/or (3) an unstable medical condition, immune deficiency, or neoplastic disease.

Skin Testing with Egg Protein and the Influenza Vaccine

The commercial egg extract was obtained from Greer Laboratories (Lenoir, NC) and was used at all participating study sites. This glycerinated extract was provided in solution at 1:20 wt/vol and stored at 4°C for stability. The influenza vaccine was obtained annually from Parke Davis (Morris Plains, NJ), and the same vaccine lot number for the corresponding year was used at all the study sites for skin prick testing and vaccine administration. The vaccine was supplied in a liquid form in 6-mL vials (with 10-mL to 0.5-mL doses) and was stored at 4°C. The epicutaneous skin testing with both the egg extract and the influenza vaccine (full-strength) was performed in standard fashion according to previously published guidelines.[13,14] A control puncture skin test was performed with 50% glycerinated saline solution. Skin test results were considered positive when the egg and/or influenza vaccine elicited a wheal of at least 3 mm greater than the control skin test site.

Inhibition ELISA for Determining Ovalbumin Content of Vaccines

The ovalbumin/ovomucoid content of the influenza vaccine was determined by an inhibition ELISA as previously described.[9] In brief, the inhibition step used a goat anti-ovalbumin polyclonal antibody plus standard serial dilutions of either ovalbumin/ovomucoid or the influenza vaccine. Three commercial preparations of the influenza vaccine were investigated including those available from Parke Davis, Connaught (Swiftwater, Pa), and Wyeth-Ayerst (Philadelphia, Pa). The solid-phase step included the standard curve dilutions mentioned above plus ovalbumin/ovomucoid coated to the plate (1 μg/mL) and peroxidase-conjugated rabbit anti-goat polyclonal antibody.

Blinded Oral Challenges with Egg Protein

As described previously, subjects who participated in the blinded oral challenges with egg were fed egg protein in a blinded or camouflaged fashion over a period of approximately 1 hour.[15,16] Each subject was observed for approximately 2 hours after the actual challenge for signs and symptoms of an allergic reaction. These included any of the following: vomiting, diarrhea, complaints of abdominal pain, coughing, wheezing, sneezing, rhinitis (clear nasal discharge), skin rashes (hives or eczema), itching, throat tightness, shortness of breath or on rare occasions anaphylaxis (a severe allergic reaction that may include significant respiratory symptoms and possible hypotension). All oral egg challenges were administered in a supervised clinical setting with appropriate emergency treatment materials immediately available if any adverse reactions or anaphylaxis should occur.

Table I. Safety of influenza vaccine administration to study groups

	Patients with egg allergy	Control subjects
No. of patients	83	124
Positive skin test response to egg	83	2
Positive skin test response to influenza vaccine	4	1
Systemic reaction after vaccination	None	None
Exact 95% CI for percentage who can safely receive this vaccine	95.7%, 100%	97.1%, 100%

Administration of Influenza Vaccine

Control subjects received a single immunizing, intramuscular dose of the influenza vaccine, kindly provided by Parke Davis, according to manufacturer's recommended dosage guidelines for age. Study subjects were given the influenza vaccine in 2 consecutive, intramuscular doses, 30 minutes apart. The first dose was one tenth and the second dose nine tenths of the recommended dose according to age and vaccine manufacturer's guidelines. Any immediate adverse reactions (generalized urticaria, laryngeal edema, bronchospasm, or other signs or symptoms of anaphylaxis) other than localized swelling at the injection site during the 30-minute observation period after the first dose of the vaccine was administered precluded administration of the second dose of the vaccine.

All vaccinations were administered intramuscularly in the posterior upper arm (middle third) in a physician-supervised clinical setting with appropriate emergency treatment materials immediately available. All vaccine recipients were observed for at least 1 hour after the vaccination to document any immediate, adverse reactions to the immunization. In addition, follow-up phone calls were made to each participant or his or her appropriate guardian 24 and 48 hours after the immunization to assess delayed adverse reactions to the vaccine.

Adverse Side Effects from Influenza Vaccination

Adverse effects after immunization with the influenza vaccine were determined immediately after and up to 48 hours after influenza immunizations. The adverse effects that were monitored included fever, local skin swelling, systemic allergic reactions, and urgent medical visits or hospitalizations attributed to the influenza immunization. Study participants were encouraged not to receive any other vaccines for a month before and a month after administration of the influenza vaccine.

Data Analysis

All data were centrally analyzed by the Biostatistics Unit at the University of Arkansas for Medical Sciences and Arkansas Children's Hospital in Little Rock. Common worksheets were used at the different clinical investigation sites to record clinical information. Exact 95% CIs were computed based on the binomial distribution by using StatXact 3 statistical software (Cytel Software, Cambridge, Mass).

RESULTS

Of the 207 subjects recruited into this multicenter clinical investigation, 83 were determined to be allergic to egg and considered study subjects (median age, 3 years; range, 1 to 46 years); 124 served as non-egg-allergic, healthy control subjects (median age, 37.5 years; range, 1 to 78.5 years). The skin prick test results with egg and safety of influenza vaccine administration for the 2 groups are summarized in Table I. All subjects with egg allergy had a clinical history compatible with egg allergy and a positive puncture skin test response to the commercial egg extract. Of these, 27 study subjects had a convincing history of anaphylactic reactions after egg ingestion, and an additional 25 subjects had their clinical histories of egg allergy confirmed with an oral challenge. Of the 124 control subjects, 109 had a negative history of egg allergy and a negative skin test response to egg; 13 had a positive history of egg allergy but a current negative skin test response to egg; the remaining 2 control subjects had equivocal histories of egg allergy and a positive skin test response to egg but underwent oral egg challenges without adverse effects.

Four (4.8%) of the 83 study subjects with egg allergy had a positive skin test response to the influenza vaccine. One of these subjects had an isolated urticarial reaction distant to the injection site after the first of 2 incremental doses of the influenza vaccine; the rash resolved without medical treatment. This patient received the second incremental dose of the vaccine 30 minutes later without experiencing any other adverse reactions. One (0.8%) of the 124 healthy control subjects had a positive skin test response to the influenza vaccine but tolerated the routine immunization without any adverse reaction.

Eight study subjects experienced some mild reactions during the 2-dose administration of the vaccine (Table II). In addition, 4 healthy control subjects had some mild reactions after the routine influenza immunization (Table II). Despite these mild adverse reactions, all of the study patients and healthy control subjects ultimately received the full immunizing dose of the influenza vaccine without experiencing any serious, systemic adverse reactions.

All of the subjects with egg allergy were safely administered this vaccine in the 2-dose protocol without experiencing significant immediate or delayed systemic reactions (exact 95% CI, 95.7% to 100%). In addition, all 124 control patients received the influenza vaccine in one routine dose (exact 95% CI, 97.1%, 100%) Current recommendations advocate a booster dose of the influenza vaccine 1 month after the initial immunization in those patients receiving the influenza vaccine for the very first time. There were 70 study subjects who were 8 years of age and under and who had egg allergy. Of these, 34 returned to receive an age-appropriate, single, full

Table II. Adverse clinical reactions reported after influenza immunization in study subjects and control subjects

Adverse reaction	Treatment	Comment
Study subjects ≤8 y of age		
1. Mild throat itching, cough, wheeze after first dose	None	Resolved; tolerated 2nd dose and booster dose
2. Small hive that resolved before 2nd dose	None	Resolved; tolerated booster dose
3. Small hive that resolved before 2nd dose	None	Resolved; tolerated booster dose
4. Delayed* emesis, mild cough, and wheeze	Neb	Resolved; tolerated booster dose
Study subjects >8 y of age		
1. Local pruritus at 18 h	None	Resolved
2. Delayed* fussiness	None	Resolved
3. Mild URI symptoms after 2-dose protocol	None	Resolved
4. Delayed* 10 mm erythema at injection site	Hydroxyzine	Resolved
Normal control subjects		
1. 50 mm erythema at injection site	None	Resolved
2. 10 mm erythema at injection site	None	Resolved
3. Emesis and fever 5 h after immunization	None	Resolved
4. Delayed* isolated urticarial reaction	Benadryl	Resolved

Neb, β_2-Agonist nebulization; *URI*, upper respiratory tract infection.
*More than 1 hour after immunization.

booster dose of the influenza vaccine 1 month after completing the 2-dose protocol, and all of these subjects tolerated the booster dose without experiencing any adverse reactions.

During the 3 years of this multicenter investigation, other commercial preparations of the influenza vaccine were found to contain significantly more egg protein as determined by an inhibition ELISA. The concentration of detectable egg protein in the Parke Davis vaccines used in this multicenter investigation ranged from 0.02 to 1.2 μg/mL. The egg protein content, determined in the Wyeth-Ayerst and Connaught preparations of the influenza vaccine, was up to 100 times higher (range, 1-42 μg/mL) than that of the Parke Davis preparation.

DISCUSSION

Previous investigations have addressed the administration of the influenza vaccine to patients with egg allergy.[3-5] Davies et al[3] recommended that patients with definite histories of egg allergy undergo skin prick testing with the influenza vaccine; if results are positive, the vaccine should not be administered. Other reports[4,5] have demonstrated that patients with egg allergy can be safely immunized if results of intradermal skin tests with the vaccine (diluted 1:100) performed before immunization are negative. Murphy et al[6] advocated a protocol of multiple, graded injections of the undiluted influenza vaccine administered to patients with egg allergy who had positive intradermal skin test responses to this vaccine. In fact, this is the current approach recommended by the American Academy of Pediatrics.[2]

Anolik et al[7] have questioned the specificity of using intradermal skin testing with a 1:100 dilution of the influenza vaccine to identify patients at risk for adverse reactions. Eight children with egg allergy who had positive skin test responses to egg and positive intradermal skin test responses to the vaccine were safely immunized with a desensitization protocol. Five of 8 healthy control subjects also had positive intradermal skin test responses to the vaccine, and all 4 patients who ultimately received the vaccine tolerated it without any adverse reactions. This study suggests that a group of eligible patients may be denied vaccination simply because of nonspecific irritant skin test reactions.

In the current multicenter clinical investigation, 4 of the patients with egg allergy had a positive skin test response to the vaccine, but they all received it in the protocol without experiencing any significant allergic reactions. This implies that the results of influenza vaccine skin testing, as suggested by previous published investigations, are unreliable. Moreover, the data pertaining to the safe administration of the influenza booster dose to study subjects provide further evidence for the safety of administering the influenza vaccine to patients with egg allergy. Administration of the vaccine in 2 doses may be no different than administration of a single dose, when the content of egg protein in the vaccine preparation can be reliably determined.

The median age of the healthy control subjects was significantly higher than that of the patients in the study group. There are 2 possible explanations for this observation. One possibility is the difficulty of recruiting pediatric control subjects who are not allergic to egg and are also clinical candidates for receiving this vaccine. The second possible explanation is the fact that patients with egg allergy typically become tolerant to this protein by their third or fourth birthday; thus, few adults have this allergic disorder. Although every study subject did not undergo an oral challenge with egg, every attempt was made to include only those patients who had a positive skin test response to egg and either a convincing history of egg allergy, a positive oral challenge response to egg, or a positive history compatible with egg allergy.

Likewise, every attempt was made to include an appropriate control group without egg allergy.

To attain a 99.9% CI of vaccination safety, approximately 8000 patients would have to receive this vaccine without any reactions in order to achieve the required statistical precision. Finding enough subjects who are both allergic to egg and clinical candidates for influenza immunization can be problematic. Alternatively, a meta-analysis, which synthesizes data from several studies, could be performed, similar to the published data evaluating the safety of routine administration of measles vaccination to patients with egg allergy.[11]

The commercial production of the influenza vaccine is undertaken on an annual basis in the hope of producing a vaccine that will protect against the relevant, predicted strains of the influenza virus; therefore, finding a vaccine with a relatively constant egg protein content can be very difficult. Over the 3 consecutive years of investigation, the concentration of detectable egg protein in the Parke Davis vaccine used in this multicenter investigation ranged from 0.02 to 1.2 µg/mL. Although this is 10-fold more than is present in a standard dose of the MMR,[9] the study subjects with egg allergy tolerated the 2-dose vaccination strategy very well, and when appropriate, a single full booster dose of the vaccine. In contrast, the other commercial preparations of the influenza vaccine with 1 to 2 log higher concentration of egg protein as compared with the commercial vaccine used in this investigation may not be tolerated as well. Future use of this protocol will depend on the determination of egg protein content in all commercially available influenza vaccine preparations. This will ensure the safest application of the information from this clinical trial and will help guide clinicians as they decide how to administer the influenza vaccine to patients with egg allergy.

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